

Project Title

IoT-Based Temperature Monitoring System Using Arduino

1. Introduction

The Internet of Things (IoT) is a modern technology that connects physical devices to the internet for communication, monitoring, and automation. IoT systems use sensors, microcontrollers, communication modules, and cloud technologies to collect and process real-time data. These systems help in reducing human effort and improving efficiency in various applications such as smart homes, industries, healthcare, agriculture, and environmental monitoring.

Temperature monitoring is one of the most common applications of IoT. Many industries and smart systems require continuous monitoring of temperature to maintain safety and proper operation. In this project, an IoT-based temperature monitoring system is designed using Arduino Uno and a TMP36 temperature sensor. The system automatically controls an LED according to temperature conditions.

When the surrounding temperature increases above a predefined value, the LED turns ON automatically. When the temperature decreases below the threshold value, the LED turns OFF. This project demonstrates the basic concept of sensor-based automation using IoT technology.

2. Aim

To design and simulate an IoT-based temperature monitoring system using Arduino Uno and TMP36 temperature sensor in Tinkercad.

3. Objectives

- To understand the concept of Internet of Things (IoT).
- To learn sensor interfacing with Arduino.
- To monitor temperature using TMP36 sensor.
- To automate LED operation based on temperature.
- To simulate the circuit using Tinkercad.
- To understand basic Arduino programming.

4. Theory

The Internet of Things (IoT) refers to a network of connected physical devices that can collect and exchange data through the internet. IoT systems consist of sensors, microcontrollers, communication devices, and software applications that work together to perform automated operations.

In this project, a temperature monitoring system is developed using Arduino Uno and TMP36 temperature sensor. The TMP36 sensor continuously senses surrounding temperature and converts it into analog voltage signals. These analog signals are sent to the Arduino through analog pin A0.

Arduino Uno acts as the main controller of the system. It reads the sensor values using its analog input pin and processes the data according to the program written in Arduino IDE. When the temperature exceeds the predefined threshold value, Arduino sends output to the LED and the LED turns ON. When the temperature falls below the threshold value, the LED turns OFF automatically.

The LED acts as an indicator for high temperature conditions. This system demonstrates automatic monitoring and control using sensor-based IoT technology.

The TMP36 sensor works on the principle of voltage variation with respect to temperature. As the temperature changes, the output voltage of the sensor also changes. Arduino converts these analog voltage signals into digital values using its built-in Analog-to-Digital Converter (ADC).

This type of IoT-based monitoring system is widely used in:

- Smart home automation
- Industrial temperature monitoring
- Fire alarm systems
- Weather monitoring systems
- Healthcare applications
- Environmental monitoring systems

The project helps in understanding the practical implementation of IoT concepts, sensor interfacing, automation, and real-time monitoring systems.

5. Components Used

Sr. No	Component Name	Description
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1	Arduino Uno	Microcontroller board used to control the system
2	TMP36 Sensor	Temperature Used to sense surrounding temperature
3	LED	Indicates high temperature condition
4	220Ω Resistor	Protects LED from excess current
5	Breadboard	Used for temporary circuit connections
6	Jumper Wires	Used for electrical connections

6. Circuit Connections

TMP36 Sensor Connections

TMP36 Pin Arduino Connection

Left Pin 5V

Middle Pin A0

Right Pin GND

LED Connections

LED Pin Arduino Connection

Positive Leg (+) Pin 13

Negative Leg (-) Resistor → GND

7. Arduino Program

```
int tempPin = A0;
```

```
int ledPin = 13;
```

```
int value;
```

```
void setup()
```

```
{
```

```
pinMode(ledPin, OUTPUT);
```

```
Serial.begin(9600);
```

```
}
```

```
void loop()
```

```
{
```

```
value = analogRead(tempPin);
```

```
Serial.println(value);

if(value > 150)
{
digitalWrite(ledPin, HIGH);
}
else
{
digitalWrite(ledPin, LOW);
}

delay(100);
}
```

8. Code Explanation

- `tempPin = A0` defines the analog pin connected to the temperature sensor.
- `ledPin = 13` defines the LED output pin.
- `analogRead(tempPin)` reads the analog value from TMP36 sensor.
- `Serial.println(value)` displays sensor values in Serial Monitor.
- If the sensor value becomes greater than 150, Arduino turns ON the LED.
- If the value is less than 150, Arduino turns OFF the LED.
- `delay(100)` provides a small delay for stable readings.

9. Working Principle

1. The TMP36 sensor continuously senses surrounding temperature.
2. The sensor converts temperature into analog voltage signals.
3. Arduino reads these analog values through pin A0.
4. The program compares the sensor value with the threshold value.
5. If temperature increases above the threshold, Arduino sends HIGH signal to LED.
6. The LED turns ON indicating high temperature.
7. When temperature decreases below the threshold, LED turns OFF automatically.
8. Sensor values can also be viewed using Serial Monitor.

10. Applications

- Smart home automation systems
- Fire warning systems
- Industrial monitoring systems
- Weather monitoring applications
- Healthcare monitoring systems
- Environmental monitoring systems
- Server room temperature monitoring

11. Advantages

- Simple and easy to implement
- Low cost system
- Automatic monitoring and control
- Low power consumption
- Real-time temperature sensing
- Useful for smart IoT applications

12. Limitations

- Limited range of operation
- Requires continuous power supply
- Sensor accuracy may vary slightly
- Suitable mainly for small-scale applications

13. Result

The IoT-based temperature monitoring system was successfully simulated in Tinkercad. The TMP36 sensor successfully detected changes in temperature and Arduino controlled the LED according to the programmed condition. The LED turned ON when the temperature exceeded the threshold value and turned OFF when temperature decreased. The project demonstrated successful implementation of temperature sensing and automatic control using Arduino and IoT concepts.

14. Conclusion

The project successfully demonstrated the design and simulation of an IoT-based temperature monitoring system using Arduino Uno and TMP36 temperature sensor. The system automatically monitored surrounding temperature and controlled an LED based on temperature conditions.

This project helped in understanding the practical implementation of IoT technology, sensor interfacing, Arduino programming, and automation concepts. Such systems are widely used in smart homes, industries, healthcare systems, and environmental monitoring applications.

The project also showed how IoT can improve automation, safety, and efficiency in real-world systems.